ANSWER KEY & MARKING SCHEME SET A

Chapters 5 & 6: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** puberty	onset of adolescence
2	**(d)** Growth of facial hair	moustache and beard
3	**(b)** 28–30 days	may vary 21–35
4	**(c)** a myth with no scientific basis	should be discouraged
5	**(b)** iron	helps haemoglobin
6	**(c)** conduction	particles pass energy
7	**(d)** Wood	used as handles
8	**(b)** conduction	heat flows along the rod
9	**(c)** from land to sea at night	sea stays warmer
10	**(d)** more than light-coloured objects	good absorbers

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Milk, curd; spinach, raisins (2).	2
12	Harassing others with digital devices (1); inform parents/teachers (1).	1+1
13	Seeping of surface water through soil and rocks (2).	2
14	Underground layers of rocks and sediments storing water (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Influence of friends (1.5); make thoughtful decisions and seek advice (1.5).	1½+1½
16	Thermocol is a poor conductor of heat (3).	3
17	Growth and repair (1.5); concentration and mood (1.5).	1½+1½
	OR: Needed for hormones (1.5); deficiency causes goitre (1.5).	1½+1½
18	Convection currents distribute heat (3).	3
	OR: Measure body temperature (1.5); shake before use, wash after use (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Blood and bones (2). (ii) Spinach, milk (1). (iii) Anaemia, weak bones (2).	2+1+2
	OR: (i) Yes (1). (ii) Hormonal changes (1). (iii) Rest, talk, hobbies (1). (iv) If it persists, talk to a counsellor (2).	1+1+1+2
20	(i) Reduces all three modes (2). (ii) Double wall, vacuum, stopper (2). (iii) Reflect heat (1).	2+2+1
	OR: (i) Metal (1). (ii) Conducts heat well (1). (iii) Conduction (1). (iv) Ice boxes made of thermocol (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 5 & 6: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** wrapped in newspaper and put in a dustbin	community health
2	**(d)** seek help from parents or teachers	handle wisely
3	**(a)** No, the timing varies from person to person	every individual is different
4	**(d)** think independently and say no	make thoughtful decisions
5	**(b)** biodegradable pads at affordable prices	through Jan Aushadhi Kendras
6	**(c)** condensation	cooling
7	**(d)** the upper level of groundwater	top of saturated zone
8	**(d)** 37 °C	98.6 °F
9	**(c)** conduction	through solid
10	**(d)** less dense	density decreases

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Builds stamina and improves mood (2).	2
12	Every individual grows at their own pace (2).	2
13	Hot air is lighter and moves upward by convection (2).	2
14	Evaporation, condensation, precipitation (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Chemical messengers trigger growth and change (3).	3
16	Heats water for evaporation and powers winds (3).	3
17	Do not respond (1); save evidence (1); inform elders (1).	1+1+1
	OR: Sports, counselling, clean toilets, awareness (3).	3
18	Warmth from the Sun, heat from a bonfire, microwave (3).	3
	OR: Particles far apart (1.5); trapped air keeps heat in (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Awareness, free or cheap pads, health (3). (ii) Improve health and access (2).	3+2
	OR: (i) Desire to belong (1). (ii) Positive habits or harmful ones (2). (iii) Think, say no, seek guidance (2).	1+2+2
20	(i) Difference (2). (ii) Solids; fluids (1). (iii) Rod heating; boiling water (2).	2+1+2
	OR: (i) Form of energy (1). (ii) Temperature is hotness, heat is energy (2). (iii) Thermometer (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 5 & 6: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Adam's apple	growth of the voice box
2	**(d)** Growth of hair in armpits	both sexes
3	**(c)** 3 to 7 days	bleeding phase
4	**(d)** the body is growing rapidly	growth spurt
5	**(c)** a blood-related health problem	needed by the body
6	**(b)** convection	movement of the fluid
7	**(d)** poor conductors of heat	do not burn hand
8	**(c)** rises and cooler fluid sinks	density difference
9	**(a)** day	differences in specific heat
10	**(d)** reflect more heat	keep us cooler

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Daily bathing and clean clothes (2).	2
12	Menstrual Hygiene Scheme or Suvidha initiative (2).	2
13	Good: copper, aluminium; poor: wood, plastic (2).	2
14	Hot air rises, warming the room by convection (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Timing and extent of changes vary (3).	3
16	More heat gives energy to ice (3).	3
17	Variations are normal (3).	3
	OR: Balanced diet, exercise, sleep, hygiene (3).	3
18	Metal is a conductor, wood is an insulator (3).	3
	OR: Convection currents and radiation (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Needs nutrients for growth (1). (ii) Iron, protein (1). (iii) Eat balanced meals (1). (iv) Doctor/dietitian (2).	1+1+1+2
	OR: (i) Facial hair, voice; breast development (2). (ii) Not involved in reproduction directly (2). (iii) Reproductive organs (1).	2+2+1
20	(i) Black (1). (ii) Absorbs more (1). (iii) Radiation (1). (iv) Solar cooker (2).	1+1+1+2
	OR: (i) Unequal heating (2). (ii) Warm air rises (1). (iii) Carry clouds (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET D

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Rashtriya Kishor Swasthya Karyakram	RKSK
2	**(a)** normal emotional changes	due to changes in the body
3	**(a)** both boys and girls	rapid growth
4	**(c)** Think before posting	dos and don’ts
5	**(a)** regular bathing and clean clothes	prevents infection
6	**(a)** precipitation	returns to Earth
7	**(d)** recharge groundwater	conserves water
8	**(b)** Radiation	e.g. Sun to Earth
9	**(c)** speeds up evaporation of sweat	removes heat
10	**(c)** conduction	contact

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rapid changes in feelings due to hormonal and emotional changes (2).	2
12	To protect privacy and avoid harm (2).	2
13	Land cools faster; air over the warmer sea rises (2).	2
14	Metal conducts heat away from the body (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Harm growth and health (3).	3
16	Infiltration fills aquifers (3).	3
17	Plan, rest, talk to teachers, exercise (3).	3
	OR: Guide, support and counsel (3).	3
18	Black absorbs heat (3).	3
	OR: Water vapour in air cools and condenses (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Safe and respectful interaction (2). (ii) Use for learning, protect privacy (2). (iii) Think before posting; avoid strangers (1).	2+2+1
	OR: (i) Diet, hygiene, exercise, sleep, social balance (3). (ii) Explanation (2).	3+2
20	(i) Clouds, rain (2). (ii) Droughts, salinity (2). (iii) Limited resource (1).	2+2+1
	OR: (i) Keep heat in or out (2). (ii) Wool, plastic, wood, thermocol, rubber (2). (iii) Cannot transfer heat when needed (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.